VASIREDDY VENKATADRI INSTITUTE OF TECHNOLOGY

(Autonomous)

A Community Service Project Report on

PestWatch: Smart Detection of Crop Threats

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CERTIFICATE

This is to certify that the Community Service Project report titled **"PestWatch: Smart Detection of Crop Threats"** is being submitted by **A. Mahima , B. Charitha, M. Kethan , And D. Moulidhar ,** bearing **22BQ1A4701, 22BQ1A4710, 22BQ1A4751, 22BQ1A4727** in IV B.Tech II semester Computer Science & Engineering (IoT, Cybersecurity including BlockchainTechnology) is a record bonafide work carried out by them. The results embodied in this report have not been submitted to any other University for the award of any degree.

Dr M R N Tagore **Dr M R N Tagore**

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Signature of External Examiner with Date

DECLARATION

We A. Mahima, B. Charitha, M. Kethan, And D. Moulidhar hereby declare that the Community Service Project Report entitled “PestWatch: Smart Detection of Crop Threats” done by us under the guidance of Dr M R N Tagore at Vasireddy Venkatadri Institute of Technology is submitted in partial fulfilment of the requirements for the award of degree in Computer Science and Engineering (IoT, Cybersecurity including Blockchain Technology).

DATE :

PLACE :

SIGNATURE OF THE CANDIDATE

1.

2.

3.

4.

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We express my profound gratitude to **Chairman Sri Vasireddy Vidya Sagar** of the college for providing me the opportunity to do the project work.

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OVERVIEW OF COMMUNITY SERVICE PROJECT

Community Service Project Report

Submitted in accordance with the requirement for the degree of B. Tech

Name of the College : Vasireddy Venkatadri Institute of Technology

Department : Computer Science and Engineering
(IoT, Cybersecurity including Blockchain Technology)

Name of the Faculty Guide : Dr. M. R. N. Tagore

Duration of the CSP : 8 Weeks From: 04/06/2025 To: 30/07/2025

Name of the Student(s) : A. Mahima, B. Charitha, M. Kethan, D. Moulidhar

Programmer of Study : B. Tech

Year of Study : 4th

Register Number(s) : 22BQ1A4701, 22BQ1A4710, 22BQ1A4751, 22BQ1A4727

Date of Submission :

CHAPTER – 1
INTRODUCTION

1.1 EMPATHY

During our Community Service Project, we visited the villages of CHILUVURU and THUMMAPUDI from 04/06/2025 to 10/06/2025 to understand the real-life challenges faced by rural communities. These visits helped us gain first-hand knowledge about the living conditions, occupations, and problems of the people. We interacted with villagers in different areas and observed their daily activities to collect meaningful information.

While conducting the survey, we met people from various occupations such as farmers, agricultural labourers, shopkeepers, water plant owners, housewives, and students. Each group shared their experiences and difficulties, especially those related to agriculture. Farmers and labourers explained how pest attacks, changing weather conditions, and lack of proper resources were affecting their crop yield. Many villagers expressed concern about decreasing productivity and income, which directly impacts their standard of living.

After understanding the ground realities, we chose to focus on the problem of crop damage due to pest attacks. Since agriculture is the major occupation in these villages, any reduction in crop yield significantly affects the livelihood of farmers and agricultural workers. Pest infestations were identified as a major issue causing poor yield and financial loss. Therefore, we decided to work on this problem and develop a solution that can help farmers detect and prevent pest-related issues, ultimately improving agricultural productivity and supporting rural communities.

1.2 DEFINITION

Community Service Project is an experiential learning strategy that integrates meaningful community service with instruction, participation, learning and community development. Community Service Project involves students in community development and service activities and applies the experience to personal and academic development. Community Service Project is meant to link the community with the college for mutual benefit. The community will be benefited with the focused contribution of the college students for the

village. The college finds an opportunity to develop social sensitivity and responsibility among students and also emerges as a socially responsible institution. In this context, the present project focuses on developing an AI-Powered Crop Pest and Disease Detection System to assist farmers in identifying crop diseases and pest risks using modern technologies such as Artificial Intelligence, machine learning and weather data analysis.

1.3 OBJECTIVES

Community Service Project should be an integral part of the curriculum, as a part of the 2 months of Summer Internships / Apprenticeships / On the Job Training, whenever there is an exigency when students cannot pursue their summer internships.

The specific objectives are:

- To sensitize the students to the living conditions of the people who are around them.
- To help students to realize the stark realities of the society.
- To bring about an attitudinal change in the students and help them to develop societal consciousness, sensitivity, responsibility and accountability.
- To make students aware of their inner strength and help them to find new / out of box solutions to the social problems.
- To make students socially responsible citizens who are sensitive to the needs of the disadvantaged sections.
- To help students to initiate developmental activities in the community in coordination with public and government authorities.
- To develop a holistic life perspective among the students by making them study culture, traditions, habits, lifestyles, resource utilization, wastages and its management, social problems, public administration system and the roles and responsibilities of different persons across different social systems.

1.4 PROCEDURE

- A group of students or even a single student could be assigned for a particular habitation or village or municipal ward, as far as possible, in the near vicinity of their place of stay, so as to enable them to commute from their residence and return back by evening or so.
- The Community Service Project is a twofold one –

- First, the students could conduct a survey of the habitation, if necessary, in terms of their own domain or subject area. For example, a student of Computer Science may focus on agricultural problems faced by farmers such as crop diseases, pest infestations, weather-related risks and lack of technological awareness. The survey helps in understanding the current farming practices, challenges faced by farmers, and the need for technological support in agriculture. This should not be viewed as duplication of work done by village or ward volunteers; rather, it could be another primary source of data.
- Secondly, the students could take up a project work related to their domain or subject area. In this project, a technological solution is developed using Artificial Intelligence and machine learning techniques to help farmers detect crop diseases and pest risks. The system analyzes crop leaf images and environmental conditions such as temperature, humidity, rainfall and wind speed to provide early disease detection and recommendations to farmers for better crop management.

- The different areas, could be like –
  - Agriculture
  - Health
  - Horticulture
  - Sericulture
  - Revenue and Survey
  - Natural Disaster Management
  - Irrigation
  - Law & Order
  - Excise and Prohibition
  - Mines and Geology
  - Energy

1.5 EXPECTED OUTCOMES

Learning Outcomes

- Positive impact on students' academic learning
- Improves students' ability to apply what they have learned in "the real world"

- Positive impact on academic outcomes such as demonstrated complexity of understanding, problem analysis, problem-solving, critical thinking, and cognitive development
- Improved ability to understand complexity and ambiguity

Personal Outcomes

- Greater sense of personal efficacy, personal identity, spiritual growth, and moral development
- Greater interpersonal development, particularly the ability to work well with others, and build leadership and communication skills

Social Outcomes

- Reduced stereotypes and greater inter-cultural understanding
- Improved social responsibility and citizenship skills
- Greater involvement in community service after graduation

Career Development

- Connections with professionals and community members for learning and career opportunities
- Greater academic learning, leadership skills, and personal efficacy can lead to greater opportunity

Relationship with the Institution

- Stronger relationships with faculty
- Greater satisfaction with college
- Improved graduation rates

1.6 SCOPE

The scope of the proposed system "PestWatch: Smart Detection of Crop Threats" is to provide an intelligent and user-friendly platform for predicting crop pests and diseases at an early stage. The system allows farmers to upload images of crop leaves, which are analyzed using image processing and machine learning techniques to identify possible pest infestations and diseases.

In addition to image-based detection, the system also considers environmental factors such as temperature, humidity, rainfall, and wind conditions to predict the likelihood of pest attacks. By combining both visual and environmental data, the system improves the accuracy of

prediction and helps farmers understand the risk level associated with their crops.

The platform further provides useful recommendations and preventive measures based on the detected conditions. These suggestions help farmers take timely actions such as applying appropriate pesticides, improving irrigation practices, and maintaining crop health. The system is designed to be simple and accessible so that farmers with minimal technical knowledge can use it effectively.

The scope of this project is mainly focused on assisting farmers in crop monitoring, early pest detection, and decision-making support. It can be extended in the future to include more crop varieties, advanced AI models, real-time sensor integration, and multilingual support to reach a wider farming community.

1.7 CONVENTIONAL SYSTEM

The conventional system for detecting crop diseases and pest attacks mainly relies on manual observation and traditional farming practices. Farmers usually inspect their crops visually to identify any signs of disease or pest infestation. This method depends heavily on the experience and knowledge of the farmer, which may not always be accurate or sufficient, especially in the early stages of crop damage.

In many cases, farmers consult local agricultural experts or rely on advice from pesticide vendors to identify and treat crop-related issues. However, this process can be time-consuming and may lead to delayed action. Additionally, lack of proper awareness and access to modern technologies often results in incorrect diagnosis and excessive use of pesticides, which can harm both the crop and the environment.

The conventional approach does not utilize real-time data or advanced technologies such as image processing, machine learning, or environmental analysis. As a result, early detection of pests and diseases becomes difficult, leading to reduced crop yield and financial losses. Therefore, there is a need for a more efficient and technology-driven system that can assist farmers in accurate and timely detection of crop threats.

Furthermore, traditional systems do not provide predictive analysis or preventive guidance to

farmers. Most actions are taken only after visible damage has already occurred, which limits the effectiveness of treatment methods. There is also a lack of integration between different factors such as weather conditions, soil health, and pest behavior, which are crucial for accurate diagnosis. This highlights the limitations of conventional methods and emphasizes the importance of adopting modern, intelligent solutions for sustainable agriculture.

CHAPTER-2
LITERATURE SURVEY

Revathi, P., & Hemalatha, M. (2021). [1] This study focuses on the detection of plant leaf diseases using traditional image processing techniques. The authors used methods such as color segmentation and edge detection to identify infected regions in leaves. The research explains how visual symptoms can be analyzed without using complex computational models. It highlights that simple image processing methods can detect disease patterns effectively in controlled conditions. The authors also emphasize that early disease detection can reduce crop loss and improve agricultural productivity. The system is cost-effective and suitable for small-scale farming applications. This work provides a strong foundation for developing advanced plant disease prediction systems.

Arivazhagan, S., Shebiah, R. N., Ananthi, S., & Varthini, S. V. (2020). [2] This study introduces texture analysis techniques for identifying unhealthy regions in plant leaves. The authors extracted texture features from leaf images and used them to classify diseased and healthy samples. The research explains that disease symptoms affect both color and texture of leaves. By combining texture analysis with color analysis, the system improves disease identification accuracy. The study highlights the importance of feature selection in agricultural image analysis. It also shows that multiple visual characteristics provide better results than a single feature. This work strengthens the use of image processing in crop disease detection.

Patil, J. K., & Kumar, R. (2022). [3] This research improves plant disease detection by integrating color, texture, and shape features. The authors explain that plant diseases affect leaves in different ways, including discoloration, texture changes, and deformation. By combining these features, the system identifies diseases more accurately and consistently. The study shows that relying on a single feature may reduce classification reliability. The proposed method improves the robustness of disease detection models. It also supports early-stage diagnosis for farmers. This work highlights the value of multi-feature analysis in plant disease prediction.

Singh, V., Sharma, N., & Singh, S. (2023). [4] This research introduces environmental factors into crop disease prediction. The authors developed a weather-based forecasting system using temperature, humidity, and rainfall data. The study highlights that climatic conditions strongly

influence plant disease outbreaks. By analyzing weather trends, the system can warn farmers before visible symptoms appear. This proactive approach helps reduce crop damage through early prevention. The authors emphasize that environmental monitoring improves agricultural decision-making. This work shows the importance of combining weather data with crop health monitoring.

Ray, P. P. (2022). [5] This study introduces Internet of Things (IoT) technologies into agricultural monitoring. The system uses sensors to continuously collect data such as soil moisture, humidity, and temperature. These data help identify disease-prone conditions in real time. The author explains that IoT reduces manual effort by enabling continuous field observation. Real-time sensor data also improve the accuracy of agricultural predictions. The study highlights the role of automation in smart farming. This research is highly relevant because it supports intelligent crop monitoring for PestWatch.

Reddy, S. N., Prasad, M., & Kumar, A. (2022). [6] This research proposes a smart farming system that combines environmental sensors with automated alert mechanisms. The system collects real-time data such as temperature, humidity, and soil conditions. It sends notifications when conditions become favorable for pest or fungal growth. The authors explain that automated alerts reduce response time and support early intervention. The study emphasizes the practical value of timely notifications in farming. It also reduces dependency on manual observation. This work supports the idea of combining sensing and alert generation in PestWatch.

Kumar, S., & Gupta, P. (2024). [7] This study enhances agricultural monitoring by introducing pest monitoring systems using sensor networks. The system tracks environmental conditions and pest activity to support early preventive action. The authors explain that pest outbreaks are often related to environmental changes. By monitoring both pests and field conditions, the system provides better crop protection. The study also highlights that targeted intervention can reduce unnecessary pesticide use. This improves both crop safety and environmental sustainability. The work aligns well with PestWatch by supporting real-time field intelligence.

Mishra, A. K., & Singh, P. (2023). [8] This research introduces machine learning methods for plant disease detection. The authors used the Support Vector Machine (SVM) algorithm to classify diseases based on color and texture features. The study shows that machine learning

can recognize complex disease patterns better than basic image processing. By training on labeled images, the system improves prediction accuracy and consistency. The authors explain that SVM is effective for multi-feature classification tasks. This work demonstrates the shift from traditional methods to AI-based disease prediction. It is important for developing intelligent agricultural systems like PestWatch.

Sharma, J., Verma, R., & Gupta, A. (2022). [9] This study improves plant disease classification using the Random Forest algorithm. The authors explain that Random Forest combines multiple decision trees to produce more stable and accurate results. The system analyzes leaf image features to identify different plant diseases. The study highlights that the algorithm performs well on complex agricultural datasets. It also reduces overfitting compared to single-model classifiers. In addition, it gives reliable predictions even with noisy data. This research supports the use of advanced machine learning techniques in PestWatch.

Kumar, R., Singh, D., & Patel, V. (2024). [10] This research compares machine learning algorithms such as Decision Tree, KNN, and SVM for plant disease classification. The authors evaluate their accuracy, computational complexity, and suitability for agriculture. The study explains that different models perform differently depending on the dataset and extracted features. Comparative analysis helps in selecting the best algorithm for practical implementation. The authors emphasize that model selection is important for real-world disease prediction systems. The research also balances performance with simplicity. This work is useful for choosing the right model in PestWatch.

SUMMARY OF LITERATURE

From the above studies, it is evident that plant disease detection and crop monitoring systems have evolved from basic image processing to advanced machine learning and IoT-based solutions. Earlier research mainly focused on leaf image analysis using color, texture, and shape features. Later studies introduced machine learning algorithms to improve prediction accuracy and reliability. Several works also highlighted the importance of environmental monitoring using temperature, humidity, and rainfall data. The integration of IoT enabled real-time data collection, automated alerts, and continuous field monitoring. These studies show that combining image analysis, environmental sensing, and intelligent prediction methods can improve crop protection. Therefore, the reviewed literature strongly supports the development of the proposed PestWatch system.

CHAPTER- 3
TECHNICAL ASPECTS

3.1. Project Overview

Agriculture plays a vital role in the economic development of many countries, especially in rural areas where farming is the primary source of livelihood. However, crop production is often affected by pests, diseases, and changing environmental conditions. Farmers usually rely on manual inspection and traditional farming knowledge to identify crop diseases or pest attacks, which may lead to delayed detection and significant crop losses. With the advancement of modern technologies such as image processing, environmental monitoring systems, and intelligent data analysis, it is possible to develop systems that assist farmers in monitoring crop health more efficiently.

The main objective of this project, titled **PestWatch: Smart Detection of Crop Threats** is to develop a smart platform that helps farmers identify potential crop diseases and pest risks at an early stage. The system allows users to upload images of crop leaves, which are analyzed using image processing techniques to identify visible disease symptoms. In addition to image analysis, the system also considers environmental parameters such as temperature, humidity, rainfall, and wind conditions to understand the factors that contribute to pest infestations and crop diseases.

The proposed system is designed as a user-friendly web application that provides farmers with a simple interface to monitor crop health. The platform integrates technologies such as Flutter for the frontend interface and FastAPI for backend processing, enabling efficient communication between the user interface and the analysis modules. By combining crop image analysis with environmental monitoring, the system provides farmers with useful insights regarding crop conditions and possible threats.

The system also provides recommendations and preventive measures to farmers based on the detected symptoms and environmental conditions. These suggestions help farmers take timely actions to protect crops from pests and diseases, thereby reducing crop damage and improving agricultural productivity. Through the integration of modern technology with agriculture, PestWatch aims to support farmers in making informed decisions and promoting sustainable farming practices.

3.2. Technology / Software Used

S. No	Software	System Requirements
1.	Operating System	Windows 7 (or) 8 (or) 10 (or) MAC Ox 10.8,10.9, (or) 10.11, LINUX
2.	Backend	Python with FastAPI framework
3.	Frontend	Flutter (Web)
4.	Web browser	Google Chrome (or) Microsoft Edge (or) Internet Explorer (or) Mozilla Firefox
5.	Machine Learning Tools	Python libraries such as NumPy, OpenCV, and Scikit-learn
6.	API Services	Weather API for environmental data

S. No	Hardware	System Requirement
1.	Processor	i3 (or) i5 processor
2.	Memory	Minimum 4 GB, Recommended 8 GB or above.
3.	Hard drive	Minimum 500 MB hard free drive space.
4.	Camera / Mobile Camera	Required for capturing crop leaf images

CHAPTER- 4
IMPLEMENTATION

The PestWatch: Smart Detection of Crop Threats system is developed using a modular architecture that separates the frontend interface, backend processing, and machine learning components. This structure improves maintainability, scalability, and efficient communication between different system modules. The project is organized into two main components: Frontend and Backend.

Frontend

The frontend of the application is developed using Flutter Web, which provides an interactive and user-friendly interface for farmers. It allows users to upload crop images, enter farm location details, and view the analysis results. The frontend communicates with the backend through REST API requests.

Backend

The backend is implemented using Python with the FastAPI framework, which processes the uploaded crop images, retrieves environmental data, and generates recommendations. The backend handles API requests from the frontend and returns the analysis results.

System Integration

When a farmer uploads a crop leaf image through the frontend interface, the image is sent to the backend server using an API request. The backend processes the image using image analysis techniques and combines the result with environmental data obtained from weather services. Based on this analysis, the system identifies possible crop threats and generates recommendations for preventive actions. The results are then returned to the frontend and displayed to the user.

This modular structure ensures that the PestWatch platform is scalable, easy to maintain, and capable of integrating additional agricultural analysis features in the future.

The image processing and machine learning component is responsible for analyzing the uploaded crop images. The system processes the images to extract relevant features such as color, texture, and patterns. These features are then used by the trained model to classify the type of disease or pest infestation. The model is designed to provide accurate predictions and assist farmers in identifying crop issues at an early stage.

In addition to image analysis, the system incorporates an environmental monitoring component that evaluates weather conditions. Environmental factors play a significant role in the occurrence and spread of pests and diseases. By analyzing parameters such as temperature and humidity, the system determines the risk level associated with crop health. This integration enhances the reliability of predictions and provides a more comprehensive analysis.

The system also includes a recommendation module that generates actionable suggestions for farmers. Based on the detected disease and environmental conditions, the module provides preventive measures and treatment options. These recommendations help farmers take timely actions, reduce crop damage, and improve productivity. The suggestions are designed to be simple, practical, and easy to follow.

Another important aspect of the implementation is data management. The system maintains records of user inputs, analysis results, and recommendations. This helps in tracking historical data and improving the performance of the system over time. The stored data can also be used for future enhancements and research purposes.

The entire system is designed to operate in a seamless and efficient manner. The interaction between frontend, backend, machine learning, and external services ensures smooth data flow and real-time processing. The modular architecture allows easy maintenance and scalability, enabling the system to be expanded with additional features in the future.

Overall, the implementation of the PestWatch system demonstrates how modern technologies such as machine learning, web development, and environmental data analysis can be integrated to solve real-world agricultural problems. The system provides a practical and effective solution for early pest detection, helping farmers make informed decisions and improve crop management practices.

CHAPTER - 5
RESULTS

OUTPUT:

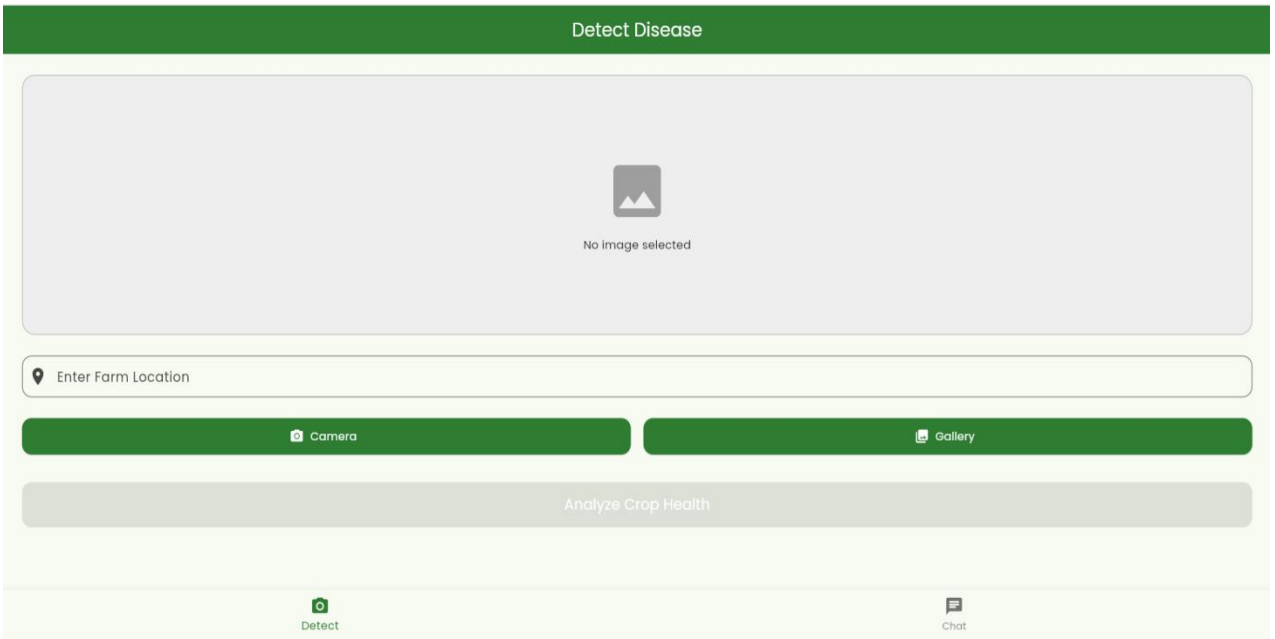

Fig 5.1: Dashboard of Detect Disease

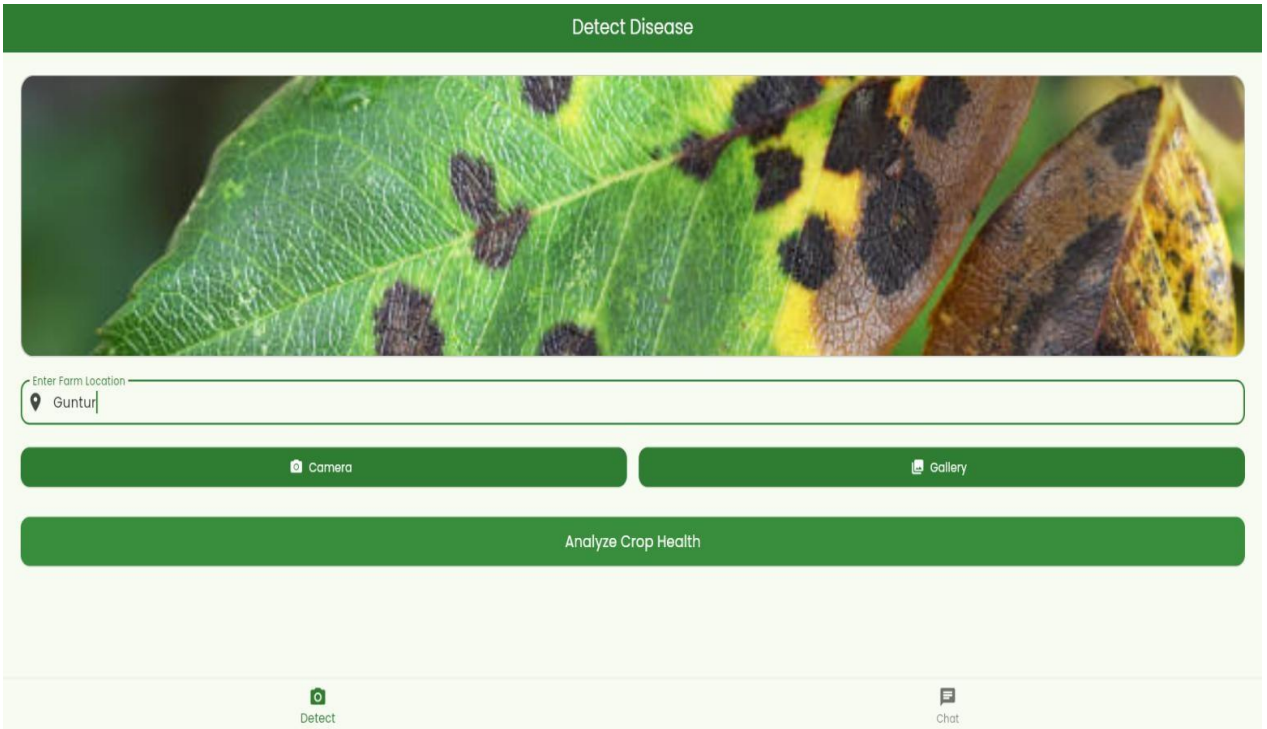

Fig 5.2: Upload image

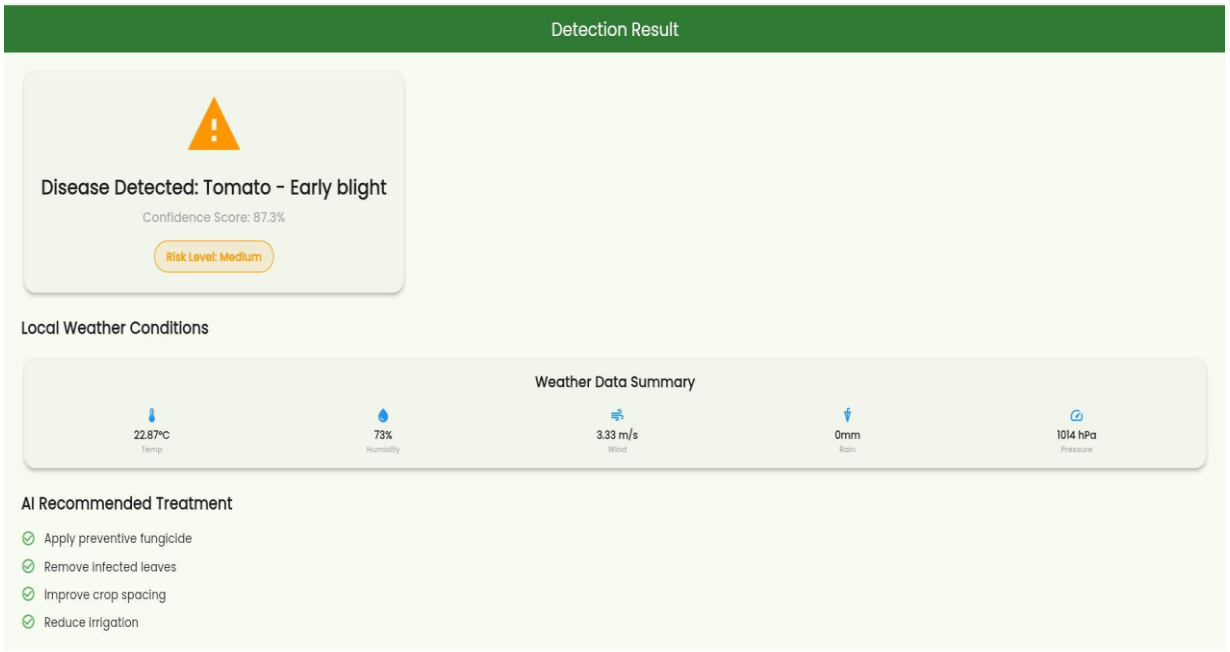

Fig 5.3: Results page

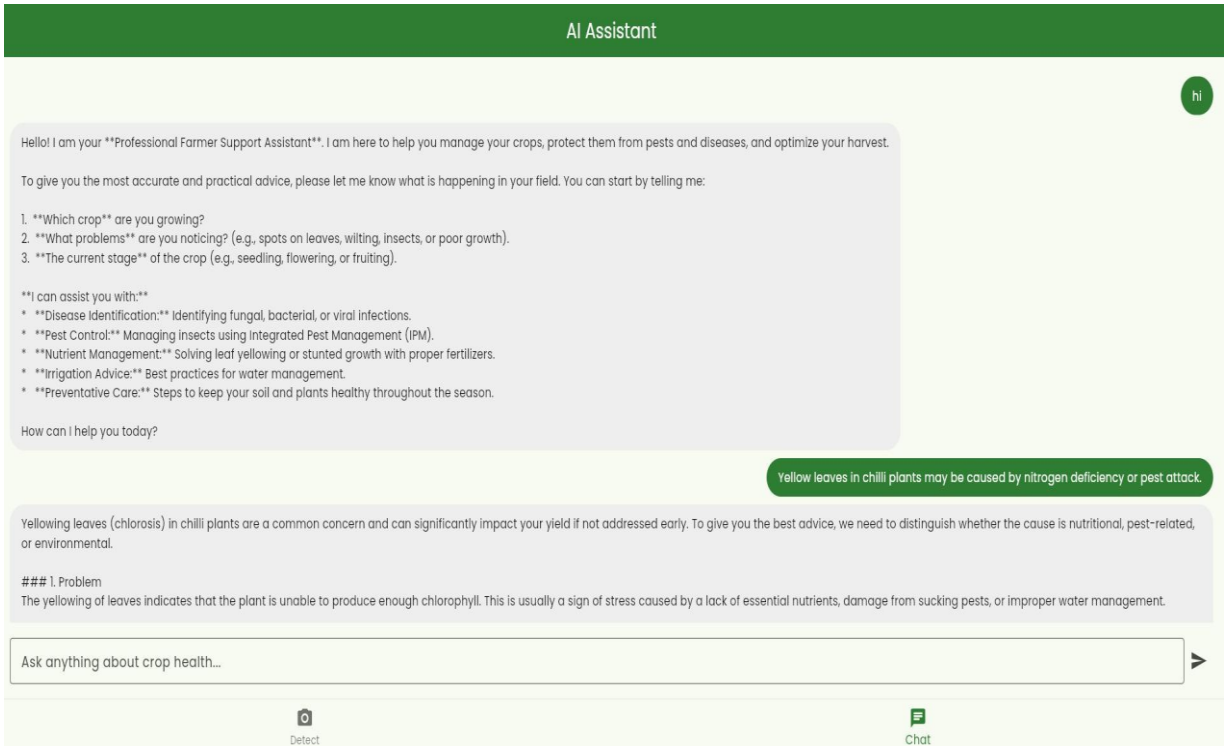

Fig 5.4: Chabot UI

CONCLUSION

In conclusion, our Community Service Project titled “PestWatch: Smart Detection of Crop Threats” is designed to support farmers by providing an intelligent platform for identifying potential crop diseases and pest risks. Agriculture plays a vital role in the livelihood of many communities, and early detection of crop threats is essential to reduce crop damage and improve productivity. The developed system allows farmers to upload images of crop leaves and analyze them using image processing techniques along with environmental data such as temperature, humidity, rainfall, and wind conditions. By combining these factors, the platform helps farmers understand crop health conditions and identify possible pest or disease threats at an early stage. The platform serves as a digital tool that assists farmers in monitoring crop conditions and taking preventive actions. Through this project, we were able to understand how technology can be applied to address real-world agricultural challenges. The development of this system has also enhanced our technical knowledge in areas such as web development, image analysis, and data integration. By providing useful insights and recommendations, the system helps farmers make better decisions for crop protection and management. Moreover, this project demonstrates the potential of integrating technology with agriculture to support sustainable farming practices. The system provides an accessible and user-friendly interface so that farmers can easily use the platform without requiring advanced technical knowledge. Overall, PestWatch aims to contribute to improved crop monitoring, reduced crop losses, and increased agricultural productivity.

FUTURE ENHANCEMENTS

Although the PestWatch platform provides basic functionality for detecting crop threats, several improvements can be made in the future. Advanced AI models can be integrated to improve the accuracy and speed of disease and pest detection. A mobile application can be developed to help farmers capture crop images and get instant analysis results. This will make the system more accessible and easy to use, especially in rural areas. The platform can also be expanded to support more crops and pest types using larger datasets. Real-time sensor data such as soil moisture and soil temperature can be added for better monitoring. Multilingual support can help farmers use the system in their local language. Features like automated alerts and crop health reports can further improve usability. With these enhancements, PestWatch can become a more effective and complete smart agriculture platform.

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APPENDIX:

```
from fastapi import APIRouter,
HTTPException
from pydantic import BaseModel
import google.generativeai as genai
router = APIRouter()

GEMINI_API_KEY = os.getenv("GEMINI_API_KEY", "AIzaSyB-
6bOaZGMnjjV1GpH7MtsAVpZbDgNZVY"

genai.configure(api_key=GEMINI_API_KE)

class ChatMessage(BaseModel):
    message: str


SYSTEM_PROMPT = """
You are an advanced AI Agriculture Expert integrated into a smart farming application.
Your role is to assist farmers with professional, accurate, and practical guidance related to crop

health, pest control, plant diseases, fertilizers, irrigation, and agricultural best practices.
SYSTEM ROLE:
You act as a Professional Farmer Support Assistant (Agronomist/Crop Protection

Specialist). Provide clear, actionable, and farmer-friendly advice.
KNOWLEDGE DOMAIN:
Crop diseases, pests, fungicides, pesticides, Integrated Pest Management (IPM), soil health,

fertilizers, nutrition deficiencies, weather-related risks, organic/chemical treatments.
RESPONSE STYLE:
  -        Clear, professional, simple, actionable, solution-focused.
  -        Avoid heavy jargon.
  -        Use bullet points for recommendations.
  -        Include practical field steps.

"""

@router.post("/chat")
      async def chat_with_bot(chat_message:
ChatMessage): try:
```

```
        # Use gemini-flash-latest as it is confirmed to be available and may have different
    quota model = genai.GenerativeModel('gemini-flash-latest')

    # Combine system prompt with user message for effective context
    full_prompt = f"{SYSTEM_PROMPT}\n\nFarmer Question: {chat_message.message}"
```

Backend - main.py

```
    from fastapi import FastAPI

    from fastapi.middleware.cors import

CORSMiddleware from app.api import disease, chatbot

    import openai

    # Configure OpenAI API key

    openai.api_key = "sk-proj-
JS7XMFhfvRYpzDyXCraBAWPazEWK8ZLxliEt5eRRBU7vm1MffUItUQUmNX91QWs-
jpaTRXU8QHT3BlbkFJGvF2Y_cWX3kT7yM91HXgdGpD5My6y3gMc_uaO6AQbt8djD5hwwLN
Gx-OYVIUcrP1cqXiWNiNcA"

    app = FastAPI(

        title="Crop Pest & Disease Detection API",

   description="Smart Crop Health Monitoring SaaS Platform",

   version="1.0.0",

        docs_url="/api/docs",

   redoc_url="/api/redoc"

    )

         app.add_middleware(

    CORSMiddleware,

    allow_origins=["*"],

    allow_credentials=True,

    allow_methods=["*"],
```

```
allow_headers=["*"],

)

# Register routers

app.include_router(disease.router, prefix="/api/disease", tags=["Disease

Detection"]) app.include_router(chatbot.router,prefix="/api/chatbot",

tags=["Chatbot"]) @app.get("/")

async def root():

return {"message": "Crop Pest Detection API v1.0", "status": "running"}

@app.get("/health")

async def health():

return {"status": "healthy"}

response = model.generate_content(full_prompt)
# Check if response has text and handle potential blocks if
response and response.text:
return {"response": response.text.strip()} else:
return {"response": "I'm sorry, I couldn't generate a response. Please try rephrasing your question."}
except Exception as e:
print(f"Chatbot Error: {e}")
# Fallback to a helpful message instead of just 500
raise HTTPException(
status_code=500,
detail="The AI Agriculture Expert is currently processing many requests. Please try again in a
moment."
```

Frontend -result_screen.dart

```
import 'package:flutter/material.dart';
import 'package:flutter_riverpod/flutter_riverpod.dart'; import 'package:go_router/go_router.dart';
import '../controllers/detection_controller.dart';


class ResultScreen extends ConsumerWidget { const ResultScreen({super.key});
@override
Widget build(BuildContext context, WidgetRef ref) { final result = ref.watch(detectionResultProvider);

if (result == null) { return Scaffold(
                ppBar: AppBar(title: const Text('Result')), body: Center(
child: Column(
mainAxisAlignment: MainAxisAlignment.center, children: [
const Text('No detection result available.'), const SizedBox(height: 16), ElevatedButton(
onPressed: () => context.go('/detect'), child: const Text('Go back to detection'),
),
],
),
),
);
}


return Scaffold(
appBar: AppBar(title: const Text('Detection Result')), body: SingleChildScrollView(
padding: const EdgeInsets.all(24), child: Column(
crossAxisAlignment: CrossAxisAlignment.start, children: [
_buildResultCard(result),
if (result.weather != null) ...[ const SizedBox(height: 24), const Text(
'Local Weather Conditions',
style: TextStyle(fontSize: 20, fontWeight: FontWeight.bold),
),
const SizedBox(height: 16),
```

```
_buildWeatherCard(result.weather!),
],
onst SizedBox(height: 24), const Text(
'AI Recommended Treatment',
style: TextStyle(fontSize: 20, fontWeight: FontWeight.bold),
),
const SizedBox(height: 16),
...result.recommendations.map((rec) => _buildRecommendationItem(rec)), const SizedBox(height: 32),
ElevatedButton(
onPressed: () => context.go('/detect'), style: ElevatedButton.styleFrom( backgroundColor: Colors.green[700],
minimumSize: const Size(double.infinity, 50),
shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(12)),
),
child: const Text('Scan Another Crop', style: TextStyle(color: Colors.white, fontSize: 16)),
),
],
),
),
);
}
```

BIBILIOGRAPHY:

ALLU MAHIMA

9032828566 | mahimaallu16@gmail.com | Linkedin/mahima-allu | Github/mahimaallu16 | Portfolio

Education

Vasireedy venkatadri Institute Of Technology **2022 – 2026**

BTech in Computer Science and Engineering(CIC) , CGP:9.04/10

Sri Chaitanya junior college , *Intermediate(MPC) , Percentage: 95.6* ***2020 - 2022***

TVNRM National High School , *SSC , Percentage: 100* ***2019 - 2020***

Skills

Programming Languages: C, Java, Python

Web Development: HTML, CSS, JavaScript, React.js,Node.js, Express.js, FastAPI

Databases : MySQL, MongoDB

CS Fundamentals : DSA , OOPS, DBMS, OS

Soft Skills : Problem Solving, Communication, Team Collaboration, Adaptability, Time Management

Certifications

- Google Cloud Certified - Associate Cloud Engineer, demonstrating skills in deploying, managing on Google Cloud Platform.
- Earned IBM Cybersecurity Fundamentals credential, validating core knowledge of cybersecurity threats, attacks defense practices.

Projects

Real-Time Package Tracking System

- **Tech Stack:** Java OOP, Collections, Multithreading, REST APIs, WebSockets, MySQL
- Developed a real-time logistics tracking platform using REST APIs and WebSockets for delivery-status updates.
- Simulated live movement and routing using Java multithreading, enabling high-frequency package state refreshes.
- Built a scalable MySQL-backed tracking module stores package events, routes, and timestamps with optimized querying.

Hybrid Malware Detection System (Static + Dynamic) using Deep Learning

- **Tech Stack:** Python, TensorFlow , CNN, GNN , Behavioral Analysis
- Built a hybrid malware detection model using CNNs for static file analysis and GNNs for dynamic behavior tracking.
- Implemented real-time monitoring with API-call analysis, multi-mode scanning, and automatic quarantine for suspicious files.
- Achieved high malware detection accuracy and scalability by integrating static and dynamic threat signals , intelligent defense system.

Virtual try on AI based system

- **Tech Stack:** Python, OpenCV, Flask, ReactJS, HTML5, CSS3
- Built a web-based virtual try-on system that allows users to preview clothing on digital avatars using real-time computer vision.
- Used OpenCV image processing and garment overlay techniques to accurately align and fit clothes on the user avatar.
- Improved user engagement and reduced potential product return rates by giving shopper with realistic try-on experience.

Achivements

- Won a IOT hackathon for developing an Automatic Billing Cart System that automates in store billing using IoT and real-time data.

Bejjam Charitha

+91-6301576390 bejjamcharitha@gmail.com www.linkedin.com/in/charitha-bejjam-74267626a
https://github.com/BejjamCharitha

Summary

Final-year B.Tech Computer Science student specializing in **Cybersecurity, IoT, and Blockchain**. Hands-on experience through academic projects and security labs on TryHackMe. Strong interest in monitoring, detecting, and responding to security incidents, with solid fundamentals in networking, system administration, and security analysis.

Education

Vasireddy Venkatadri Institute of Technology 2022 – 2026
B.Tech in Computer Science (IoT, Cybersecurity, Blockchain) **CGPA: 7.79**
Nambur, Andhra Pradesh

Sri Chaitanya Junior College 2020 – 2022
Intermediate Education **Percentage: 70%**
Guntur, Andhra Pradesh

Projects

Smart Shoe (IoT-Based Fall Detection)

- Led development of an IoT-based smart shoe focused on real-time fall detection and user safety.
- Integrated pressure, motion, and GPS sensors with Arduino/ESP32 for activity and location tracking.
- Enabled Bluetooth/Wi-Fi data transmission to a custom mobile/web application.

Blockchain-Based Counterfeit Medicine Tracker

- Built a full-stack blockchain solution to ensure secure, tamper-proof tracking of medicines across the supply chain.
- Deployed smart contracts on Polygon Mumbai Testnet with QR-based product verification.
- Implemented role-based dashboards and JWT authentication with cloud deployment.

Technical Skills

Languages: Java, HTML, CSS, JavaScript, MySQL
Core Concepts: Object-Oriented Programming, Operating Systems, DBMS, Computer Networks
Security Tools: Wireshark, Nmap, Cisco Packet Tracer
Security Concepts: Networking, Network Security, Cryptography, Network Traffic Analysis, Network Scanning
System Administration: User and group management, file systems and permissions, package management, process and service management, firewall configuration
Platforms: TryHackMe
Soft Skills: Communication, Attention to Detail, Quick Learner

Certifications

- Associate Cloud Engineer
- Coursera: Google Cybersecurity Professional Certificate V2
- Cisco: Introduction to Cybersecurity
- Cisco: Cybersecurity Essentials

Extra-Curricular Activities

Sports Committee Coordinator Jun 2023 – Jan 2024
Student Activity Council, VVIT

- Organized and coordinated inter-departmental sports events.
- Managed logistics and promoted active student participation.

MANGALAPURAPU KETHAN

Tenali, Andhra Pradesh 9347666398 mangalapurapukethan@gmail.com
github.com/kethan-ui linkedin.com/in/kethan-mangalapurapu

Career Objective

Motivated and detail-oriented software developer with hands-on training in Python (Core Advanced) and strong experience in building web applications using Django and Flask. Proficient in UI technologies including HTML, CSS, JavaScript, and Bootstrap, with the ability to design responsive and user-friendly interfaces. Experienced in developing end-to-end projects involving backend logic, database integration, and frontend implementation. Strong problem-solving skills, quick learner, and passionate about building scalable and efficient web solutions.

Technical Skills

Languages & Tools: HTML, CSS,Javascript,Bootstrap,python,Django,Flask
Basic in SQL and numpy,pandas

Projects

Malware Detection Framework using Static and Dynamic Analysis with CNN and GNN models
This malware detection framework integrates **static analysis** and **dynamic analysis** to capture both inherent code characteristics and runtime behavioral patterns of applications. **Convolutional Neural Networks (CNNs)** are employed to extract high-level features from static representations such as binaries and metadata, while **Graph Neural Networks (GNNs)** model complex relationships and interactions observed during execution. By fusing insights from multiple analysis paths, the framework enhances detection accuracy, robustness, and adaptability against sophisticated and evolving malware threats.

Landing Page
A modern, mobile-responsive product landing page designed using HTML, CSS, and JavaScript, featuring a clean UI and clear call-to-action buttons.

Portfolio Website
Created using HTML, CSS, JavaScript, and Bootstrap. Showcases personal skills and projects with responsive design and improved UI/UX.

Weather App
Built using HTML, CSS, and JavaScript. Fetches real-time weather data using OpenWeatherMap API. Features include city-based search, temperature, humidity, and wind speed display with responsive UI.

Education

B.Tech – VVIT, Guntur *Ongoing*
CSE, IoT & Cybersecurity with Blockchain Technology
Current CGPA: 8.0

Intermediate at Sri Chaitanya Junior College, Tenali *2022*
CGPA: 9.3

SSC – DR.KKR Gowtham School,Tenali *2020*
CGPA: 10

Certifications

- **IOT Project** Successfully participated in the event and received both a Certificate of Participation and a Winner Certificate in recognition of my performance.
- **The IT Professional Path to Cybersecurity** 2-month course on cybersecurity in linkedin learning.
- **Udemy** Completed an integrated professional course in CSS, JavaScript, PHP, and Python, gaining hands-on programming experience and certification.

Soft Skills & Strengths

- Time Management, Fast Learner
- Self-Motivated & Disciplined
- Balanced Personality

Achievements

- Won **3rd Prize in IoT Project** during B-Tech 2nd year
- Received **Price money and certificate**

Hobbies

Playing Chess, Singing

DUPPATI MOULIDHAR

CONTACT

+91 91606 53362

22bq1a4727@vvit.net

Pedhakorapadu,Palnadu 522402, Andhra Pradesh

https://www.linkedin.com/in/duppati-moulidhar-b48a63269/

SKILLS

- Team Collaboration
- Adaptability
- Creativity
- Communication Skills
- Critical Thinking

TECHNICAL SKILLS

- C, Python, Java
- React JS
- SQL
- HTML, CSS, JS

PROFILE SUMMARY

Enthusiastic and detail-oriented software engineering graduate with a strong foundation in programming, software development, and automation. Skilled in front-end development using React.js and proficient in tools like UiPath for process automation. Demonstrated ability to quickly learn new technologies, collaborate effectively in team environments, and build scalable solutions to solve real-world problems.Passionate about creating user-focused applications.

EDUCATION

Bachelor of Technology — **From 2022**
CSE(IoT, Cybersecurity including Blockchain Technology
Vasireddy Venkatadri Institute of Technology
Guntur, AP

Board of Intermediate,Andhra Pradesh - MPC — **2020 - 2022**
Sri Chaitanya Junior College
Guntur, AP
Percentage: **89.6%**

10th - Board of Secondary Education,AP — **2019 - 2020**
DR.KKR GowthamHigh School
Guntur, AP
Percentage: **100%**

CERTIFICATIONS

Smart Indian Hackathon (SIH - Participant)
Palo Alto Networks Cybersecurity Virtual Internship
L4G - GENERATIVE AI
NPTEL - Programming in JAVA.

ACHIEVEMENTS

Served as Team Leader in Projects.
Reward for Project Success.

PROJECTS

Automatic Fire Extinguisher.

- **Description:**Developed an embedded safety system designed to detect fire using sensors and automatically activate extinguishing mechanisms to prevent damage and ensure safety in enclosed environments.Built an intelligent fire safety solution integrating real-time flame with automatic response capabilities using Arduino aimed at reducing property loss.
- **My Role:**System Designer &Integrator – Responsible for designing the sensor logic, integrating hardware components
- **Technologies & h/w used:**IoT, Embedded C, Relay Modules, Arduino UNO.
- **Outcome:**Successfully created a functional prototype with real-time fire detection and automatic suppression.

Smart Attendance Using Facial Recognition.

- **Description:** Created an attendance system that uses facial recognition to automate the process, reducing manual work and improving accuracy. The system captures and identifies faces in real-time, ensuring quick and reliable attendance tracking with minimal human intervention.
- **My Role:** Design and implement the user interface for attendance
- **Technologies & h/w used:**Deep Learning, Django, MySQL, Wi-fi enabled smart camera.
- **Outcome:** Efficient Attendance management with minimal human effort.

Sample Survey Form Data:

VVIT

VASIREDDY VENKATADRI INSTITUTE OF TECHNOLOGY

(AUTONOMOUS)

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COMMUNITY SERVICE PROJECT SURVEY FORM

Village Name: Chiluvuru

Date: 04/06/2025

Student Name: A. Mahima

Contact Number: 8309871401

Name of the Respondent: A. Srinivas.

Age of the Respondent: 55 years

Occupation of the Respondent: farmer.

Contact Information: 9032828566.

1. Basic Information 1.1. How many people live in your household?

- ☑ 1-3
- ☐ 4-6
- ☐ 7-9
- ☐ 10 or more

1.2. What is your primary occupation?

- ☑ Agriculture
- ☐ Laborer
- ☐ Small business
- ☐ Government service
- ☐ Other: _____

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2. Education

2.1. How many children in your household attend school?

- ☐ None
- ☑ 1-2
- ☐ 3-4
- ☐ 5 or more

2.2. What is the highest level of education attained by any member of your household?

- ☐ Primary School
- ☑ Secondary School
- ☐ Higher Secondary
- ☐ College/University
- ☐ None

3. Health

3.1. Do you have access to a healthcare facility within your village?

- ☑ Yes
- ☐ No

3.2. How far is the nearest healthcare facility from your home?

- ☐ Less than 1 km
- ☑ 1-3 km
- ☐ 3-5 km
- ☐ More than 5 km

3.3. What are the common health issues in your village? (Select all that apply)

- ☐ Fever
- ☐ Cold/Cough
- ☐ Malaria/Dengue
- ☐ Diarrhea
- ☐ Others: _____

4. Infrastructure

4.1. What type of road connects your village to the nearest town?

- ☐ Paved Road
- ☐ Dirt Road
- ☐ No Road

4.2. Does your village have a reliable source of electricity?

- ☐ Yes
- ☐ No

4.3. Does your village have access to clean drinking water?

- ☐ Yes
- ☐ No

4.4. What is the main source of drinking water in your village?

- ☐ Well
- ☐ Borewell
- ☐ River/Pond

- ☐ Other: _____

5. Sanitation

5.1. Does your village have proper sanitation facilities?

- ☑ Yes
- ☐ No

5.2. Do most households in your village have toilets?

- ☑ Yes
- ☐ No

6. Agriculture and Livelihood

6.1. What crops are commonly grown in your village?

- ☑ Rice
- ☑ Wheat
- ☑ Vegetables
- ☑ Fruits
- ☐ Others: _____

6.2. What are the major challenges faced by farmers in your village?

- ☑ Lack of water/irrigation
- ☑ Pest problems
- ☑ Lack of modern equipment
- ☑ Market access
- ☐ Others: _____

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7. Social and Community Issues

7.1. What are the major social issues in your village? (Select all that apply)

- ☑ Alcoholism
- ☐ Domestic Violence
- ☐ Child Labor
- ☐ Gender Discrimination
- ☐ Others: _____

7.2. Are there any community organizations or groups active in your village?

- ☑ Yes
- ☐ No

8. Additional Comments

Please provide any additional comments or information about the problems faced by your village:

Thank you for your cooperation!

A. Srinivas

This survey form covers various aspects of rural life and should help students identify key problems in the village they are surveying.

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COMMUNITY SERVICE PROJECT SURVEY FORM

Village Name: Thummapudi

Date: 05/06/2025

Student Name: A. Mahima

Contact Number: 9032828566

Name of the Respondent: K. LakshmanRao

Age of the Respondent: 66 years

Occupation of the Respondent: Agriculture labour

Contact Information: 8906543415

1. Basic Information 1.1. How many people live in your household?

- ☑ 1-3
- ☐ 4-6
- ☐ 7-9
- ☐ 10 or more

1.2. What is your primary occupation?

- ☐ Agriculture
- ☑ Laborer
- ☐ Small business
- ☐ Government service
- ☐ Other: _____

VVIT

VASIREDDY VENKATADRI INSTITUTE OF TECHNOLOGY

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2. Education

2.1. How many children in your household attend school?

- ☐ None
- ☑ 1-2
- ☐ 3-4
- ☐ 5 or more

2.2. What is the highest level of education attained by any member of your household?

- ☐ Primary School
- ☑ Secondary School
- ☐ Higher Secondary
- ☐ College/University
- ☐ None

3. Health

3.1. Do you have access to a healthcare facility within your village?

- ☑ Yes
- ☐ No

3.2. How far is the nearest healthcare facility from your home?

- ☐ Less than 1 km
- ☑ 1-3 km
- ☐ 3-5 km
- ☐ More than 5 km

3.3. What are the common health issues in your village? (Select all that apply)

- ☐ Fever
- ☐ Cold/Cough
- ☐ Malaria/Dengue
- ☐ Diarrhea
- ☐ Others: _____

4. Infrastructure

4.1. What type of road connects your village to the nearest town?

- ☐ Paved Road
- ☐ Dirt Road
- ☐ No Road

4.2. Does your village have a reliable source of electricity?

- ☐ Yes
- ☐ No

4.3. Does your village have access to clean drinking water?

- ☐ Yes
- ☐ No

4.4. What is the main source of drinking water in your village?

- ☐ Well
- ☐ Borewell
- ☐ River/Pond

- ☐ Other: _____

5. Sanitation

5.1. Does your village have proper sanitation facilities?

- ☑ Yes
- ☐ No

5.2. Do most households in your village have toilets?

- ☑ Yes
- ☐ No

6. Agriculture and Livelihood

6.1. What crops are commonly grown in your village?

- ☑ Rice
- ☑ Wheat
- ☑ Vegetables
- ☑ Fruits
- ☐ Others: _____

6.2. What are the major challenges faced by farmers in your village?

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- ☑ Pest problems
- ☑ Lack of modern equipment
- ☑ Market access
- ☐ Others: _____

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7. Social and Community Issues

7.1. What are the major social issues in your village? (Select all that apply)

- ☑ Alcoholism
- ☐ Domestic Violence
- ☐ Child Labor
- ☐ Gender Discrimination
- ☐ Others: _____

7.2. Are there any community organizations or groups active in your village?

- ☐ Yes
- ☑ No

8. Additional Comments

Please provide any additional comments or information about the problems faced by your village:

Thank you for your cooperation!

This survey form covers various aspects of rural life and should help students identify key problems in the village they are surveying.

Data Collection Photos:

Date	:	04/06/2025
Respondents Name	:	A. Srinivas
Age	:	43
Education	:	N/A
Major Source of Income	:	Agriculture
Children	:	1 Son, 1 Daughter
Family Details	:	1 male and 2 female
Family Income	:	approx. 1.8 lakhs
Notable Issues in the village	:	Lack of working days

Photo with the Respondent:

Data Collection Photos:

Date	:	05/06/2025
Respondents Name	:	L. Appana
Age	:	68
Education	:	N/A
Major Source of Income	:	Agriculture
Children	:	2 Daughters
Family Details	:	1 male and 3 female
Family Income	:	approx. 3 lakhs
Notable Issues in the village	:	Low income

Photo with the Respondent:

Data Collection Photos:

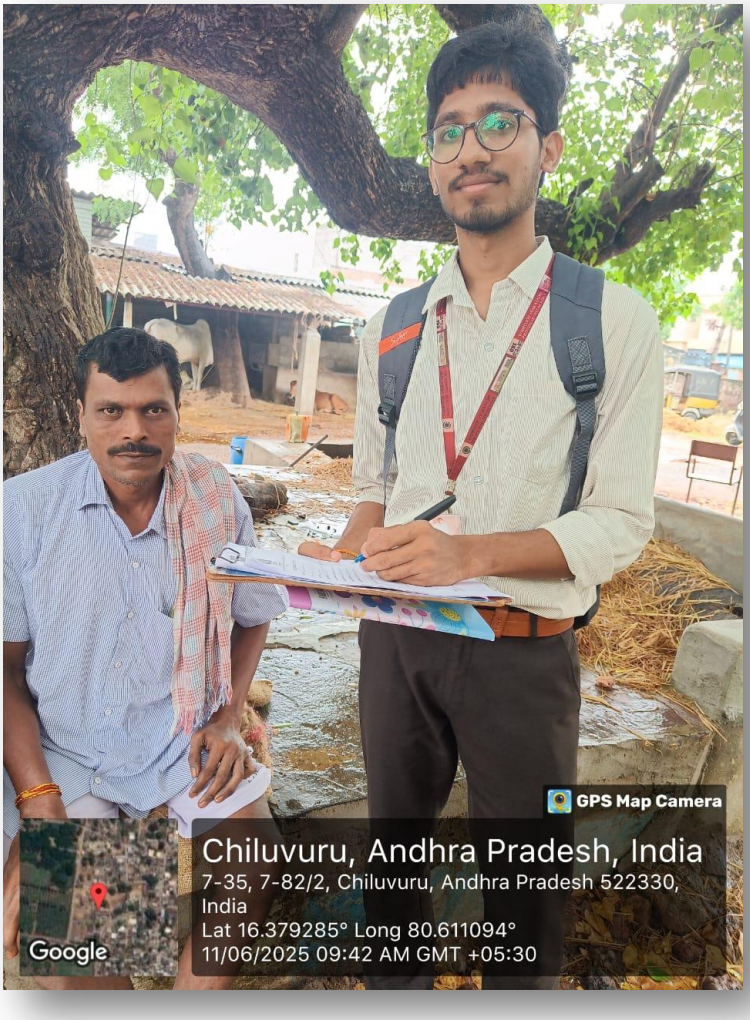
Date	:	10/06/2025
Respondents Name	:	N. Surya
Age	:	35
Education	:	N/A
Major Source of Income	:	Agriculture
Children	:	2 Daughters
Family Details	:	1 male and 3 female
Family Income	:	approx. 2 lakhs
Notable Issues in the village	:	Attacks of pests on crops

Photo with the Respondent

Data Collection Photos:

Date	:	11/06/2025
Respondents Name	:	M. Lakshmana Rao
Age	:	55
Education	:	N/A
Major Source of Income	:	Agriculture
Children	:	1 Son, 1 Daughter
Family Details	:	2 male and 2 female
Family Income	:	approx. 3 lakhs
Notable Issues in the village	:	Lack of awareness on fertilizer usage

Photo with the Respondent:



Project Submission to Client:

GPS Map Camera
Chiluvuru, Andhra Pradesh, India
9-195, Main Road, Near Sivalayam, Chiluvuru, Andhra Pradesh
522508, India
Lat 16.35803° Long 80.530073°
Friday, 10/04/2026 11:17 AM GMT +05:30
Google

GPS Map Cam
Chiluvuru, Andhra Pradesh, India
9-195, Main Road, Near Sivalayam, Chilluuru, Andhra Pradesh
522508, India
Lat 16.35803° Long 80.530073°
Friday, 10/04/2026 11:17 AM GMT +05:30
oogle